Stacked n = 83 K600 producer: probe pipeline + 16-byte slots + state fold + rank stage, end-to-end cold

2026-10-16

Stacked n = 83 K600 producer: probe pipeline + 16-byte slots + state fold + rank stage, end-to-end cold

Status (2026-10-16): measured; frozen primary rule applied; verdict `improvement`. GitHub Actions run 38862448935 (attempt 1, one `ubuntu-24.04` runner, Tier 1 instance scope) ran the frozen protocol and schedule, committed before any Tier 1 timing. Its artifact `n83-stacked-38862448935-1` is archived in `run_github_38862448935/`. On 20 new points and two rank blocks the stack lowers the **end-to-end cold** cost (rank stage + setup + one-target online) by **3.2391 [2.9809, 3.5240]** (E2E(kP, cP) / E2E(kS, cS)). See Result. The numbers under "Local validation" and in `smoke/` are diagnostics from the cloud-agent VM, not timing evidence.

Question

Main has five promoted n = 83 a = 1 K600 one-target IC improvements, each measured on its own parent and none end to end:

step	experiment	promoted on
lift_scan canonicalizer	`koblitz-n83-liftscan-20261015` (#926)	online, over B
Feistel scan order	`koblitz-n83-scanorder-20261015` (#927)	online, over B
both combined	`koblitz-n83-combined-20261015` (#929)	not promoted (`not_separated` on geomeans)
prefetched probes (pP)	`koblitz-n83-probepipe-20261015` (#931)	online 1.40x over the combined producer
two-level root table	`koblitz-n83-tablefilter-20261015` (#934)	online 1.063x over pP, almost all from the 16-byte slot (pK 1.058x)
folded S3 root index	`koblitz-n83-state-fold-20261015` (#933)	cold (setup + online) 1.84x over the combined producer; online about 5% slower; its rank stage on lift + Feistel was 2.25x cheaper per attempt (secondary)

None of them moved the n = 83 row of `docs/ic/BASELINES.md`, because none measured an end-to-end cold `S` (setup + index build + rank stage + online), and the rank stage (B's: about 750 s on the runner) dominates that cost.

Does the stack of all of them, with the rank stage run on the stack's own probe loop, lower the **end-to-end cold** cost of the current best producer (pP, whose rank stage is B's) on new targets, with every relation, scalar and factor-log table unchanged? That is an iteration under AGENTS.md "Success criterion"; a promoted result is the first n = 83 row with a measured end-to-end cold `S` after B.

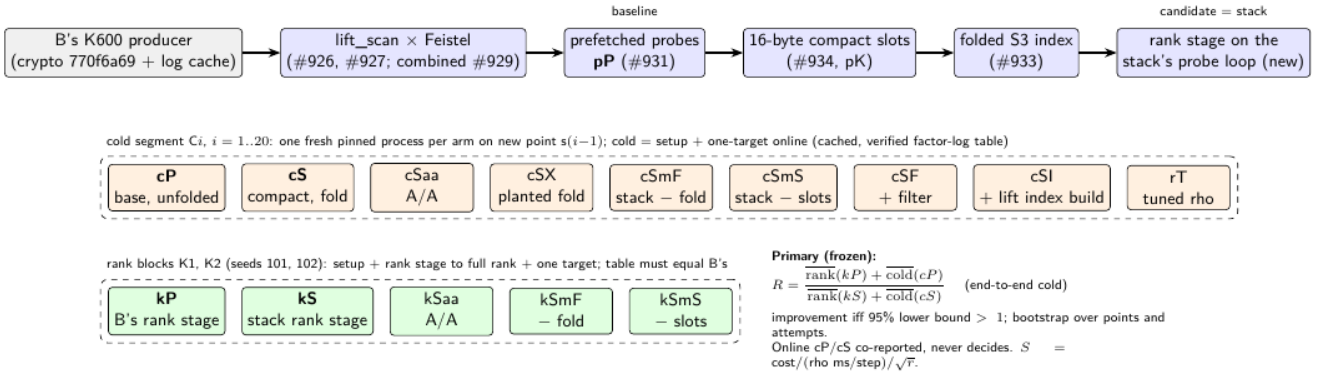


Figure 1: Design (from the frozen protocol; not a result). Source: figures/design.tex, vector: figures/design.pdf.

What is stacked

`make_online_source.py` derives the producer in two steps:

1. `make_tablefilter_source.py` is `koblitz-n83-tablefilter-20261015`'s `make_online_source.py`, verbatim (with its two verbatim parents, `make_probepipe_source.py` and `make_combined_source.py`). Its output must hash to that protocol's frozen producer digest (`70233510...`). With `KIC_ROOT_TABLE=base` and `KIC_PROBE_PIPELINE=prefetch` it is the probe-pipeline producer pP; with `KIC_ROOT_TABLE=compact` it runs the 16-byte slots.
2. Counted, exact replacements port the state fold (`koblitz-n83-state-fold-20261015`, `FOLD_CODE` and `fold_edits`) onto every root-table layout and probe loop:
   - `KIC_STATE_FOLD=1` stores 14,940,300 representatives of the 29,880,000 ordered states, fills **the selected table** (32-byte or 16-byte slots, or the filter's compact table) from them in the baseline's insertion order (capacity follows the stored count, 2^26 slots), and the folded scan now calls the layout's own probe loop (`scan_state128_select::<C, P>`) for each class at its first occurrence in the baseline order; mates are rebuilt by two rotations. `KIC_FOLD_PLANT` and `KIC_INDEX_CANON` keep the state fold's semantics.
   - **The rank stage** gets `KIC_RANK_CANON` and `KIC_RANK_ORDER` (as in the state fold) and the new `KIC_RANK_PROBE=target`, which runs the arm's own probe loop in the rank stage: the prefetch pipeline on the 32-byte table, or the compact (or filter) loop on the 16-byte table. So the stack's rank stage runs on the 16-byte table in the same process; the parent's rule that a non-base layout needs the cached factor-log table now applies only to `KIC_RANK_PROBE=off`. Each rank process records every attempt's probes and ns and `rank_relations_blake3`, a digest of every attempt's relation.
   - `KIC_INDEX_DIGEST=1` writes the state fold's index digests (logical states in index order; the table's key -> value map), read from whichever table the index holds, after the target and outside every interval.

Exactness argument. Each part is exact on its own (its parent's tests and gates): the 16-byte slot stores the same key -> value map and the same probe sequence; the pipeline keeps probe order and early return; the fold finds the baseline's first relation with fewer probes (the skipped mates made the same probes without a hit). The port changes only which loop the folded scan calls, and every loop takes a `State128`, so a folded scan over the compact table makes the compact loop's probes on the baseline's logical order. In the rank stage, the attempts are a deterministic function of the rank seed and the earlier relations; if every attempt's relation is unchanged, the rank stage's echelon, attempts and factor-log table are unchanged. The tests and the smoke below check all of this.

Left out, and why.

- **The root filter** (`KIC_ROOT_TABLE=filter`) is not in the stack: on the runner it added 1.005x on top of the 16-byte slot (run 38829449594). It is a secondary arm (`cSF`), cold only.
- **lift_scan in the index build** (`KIC_INDEX_CANON=lift`, the state fold's cF/cFI 1.33x cold secondary) was never promoted. It is a secondary arm (`cSI`), cold only; the time budget did not allow its own rank arm, and its rank stage is the stack's (it changes only the build).

Frozen Tier 1 design

One GitHub `ubuntu-24.04` runner, one pinned CPU, Tier 1 counters, one fresh process per run. Every IC arm runs lift_scan, the Feistel order and the prefetch pipeline on the online path.

Arm	Table	Fold	Rank stage	Role
`cP` / `kP`	32-byte	no	B's: full scan, left-major, sequential probes	**baseline** pP
`cS` / `kS`	16-byte	yes	lift, Feistel, the stack's compact loop	**candidate**: the stack
`cSaa` / `kSaa`	16-byte	yes	as kS	A/A
`cSX`	16-byte	yes + planted	(cold only)	planted positive control: the fold's removed work, redone and discarded
`cSmF` / `kSmF`	16-byte	no	lift, Feistel, compact loop	ablation: stack minus fold
`cSmS` / `kSmS`	32-byte	yes	lift, Feistel, prefetch pipeline	ablation: stack minus slots
`cSF`	filter + 16-byte	yes	(cold only)	optional component: 32 MiB filter
`cSI`	16-byte	yes, lift in build	(cold only)	optional component: lift_scan index build
`rT`	—	—	—	tuned rho rT770 (Fold/64/DP6), seed 870000 + point seed

- **Cold panel.** 20 new public points (`inputs/cold_targets.json`, role `stacked-cold-v1`, `make_targets.py`), checked disjoint from Q0, B's panel and every earlier n = 83 panel of the law on main (scan-order held-out 384, combined 576, probe-pipeline 384, state-fold 16, table-filter 384), all regenerated. No scalar is known for any of them. Each `C<i>` segment runs one process per cold arm on point s(i-1), shuffled.
- **Rank blocks.** `K1`, `K2` (rank seeds 101, 102): one process per rank arm, each: setup, the rank stage to full rank, linear algebra, then one target (s000, s001). Every written factor-log table must equal B's archived table, log for log.
- **Q0 block** `O1`: cP and cS on Q0, exactness against the archived Feistel identity.
- **Segment order:** O1, C1-C4, K1, C5-C12, K2, C13-C20; 300-minute guard.

Primary (pre-registered): the end-to-end cold ratio. E2E(k, c) = mean rank stage of rank arm k over its two processes + mean cold (setup + one-target online, cached table) of cold arm c over the 20 points. R = E2E(kP, cP) / E2E(kS, cS); bootstrap: points resampled (paired), and the attempts of each rank process resampled (its non-attempt time kept), 10,000 draws. **Improvement** iff the 95% lower bound of R exceeds 1; regression iff the upper bound is below 1; otherwise not separated. **Invalid** if any gate or control fails or any scheduled row is missing.

Online is co-reported, never decides: online cP/cS (means and geomeans), and the fold's online cost inside the stack (online cSmF/cS).

- **Controls.** A/A: E2E(kS, cS)/E2E(kSaa, cSaa) and cold cS/cSaa both inside [0.97, 1.03]. Planted: cold cSX/cS lower bound above 1.
- **Gates.** Every IC process: its layout, slot count (2^27 unfolded, 2^26 folded), 29,880,000 logical states, 14,940,300 stored iff folded, 29,878,182 table entries, and B's index digests. Per point: one scalar across every IC process and rT (oracle replay); one relation across every IC process; probes cP = cSmF; cS = cSaa = cSX = cSmS = cSF = cSI $\leq$ cP; cSX + audit = cP. Rank: B's table; per block, one relation digest across kS, kSaa, kSmF, kSmS; identical per-attempt probes across kS, kSaa, kSmS; kSmF $\geq$ kS per attempt. Q0: archived relation and scalar, probes 1,988,184 (cP) and 1,987,852 (cS).
- **Secondaries.** Online, cold-without-rank, setup, index and peak-RSS ratios cP/cS; ablations E2E(kSmF, cSmF)/E2E(kS, cS) and E2E(kSmS, cSmS)/E2E(kS, cS) with their online, cold and setup ratios; cS over cSF and cSI; rank-stage and per-attempt ratios kP/kS, kSmF/kS, kSmS/kS; each rank process's single-process

end-to-end cold time; `S` = cost / (rT ms/step) / sqrt(r) for end-to-end cold, cold without rank and online, per arm; delta `S` cP to cS; the end-to-end cold gap IC / rT and the online gap to rT (distance remaining, not a gate).

- **Power** (`power_check.py`, `analysis/power.json`; prior `inputs/power_prior.json` built by `make_power_prior.py` from the state fold's run 38819494806 and three measured factors). The rank stage, not the point count, limits resolution. At the minimal effect of interest, 1.15, the power is 0.88-0.93 at every size from 4 to 20 points (300 simulations; the smallest size passing 0.80 after two Monte Carlo SEs is 4); at 1.10 it is 0.55-0.63, so end-to-end effects of 10% or less are not reliably resolved by this design. The prior's expected ratio is 3.2 (pP rank stage about 757 s, cold about 42 s; stack about 228 s and 22 s). 20 points serve the per-point online ablations (power to see the fold's 5% online cost inside the stack: 0.98 at 8 points, 1.0 at 16-20).
- **Runtime estimate.** About 3 h including the build (up to about 3.5 h on a slower runner); 300-minute time guard, 350-minute job limit.

Local validation (cloud-agent VM; diagnostics, no timing claims)

See `smoke/README.md`.

Result

All gates pass: every scheduled row is present and valid (192 valid, 0 invalid, 0 skipped by the time guard; 20/20 points complete), probe identities, one relation per point across every IC process, one scalar per point across every IC process and the tuned rho (replayed in the pinned oracle), Q0 equal to the archived Feistel identity, every rank process's factor-log table equal to B's archived table, and identical rank relations and per-attempt probes across layouts. Both controls pass. `verify_receipt.py` re-run on the archive reproduces `panel/result.json`, `panel/result_table.md` and `verify-stdout.json` byte for byte. The verifier flags one contention note on two rows: major page faults on serial 2 (`cS`) and serial 12 (`cSaa`); both rows are valid and stay in.

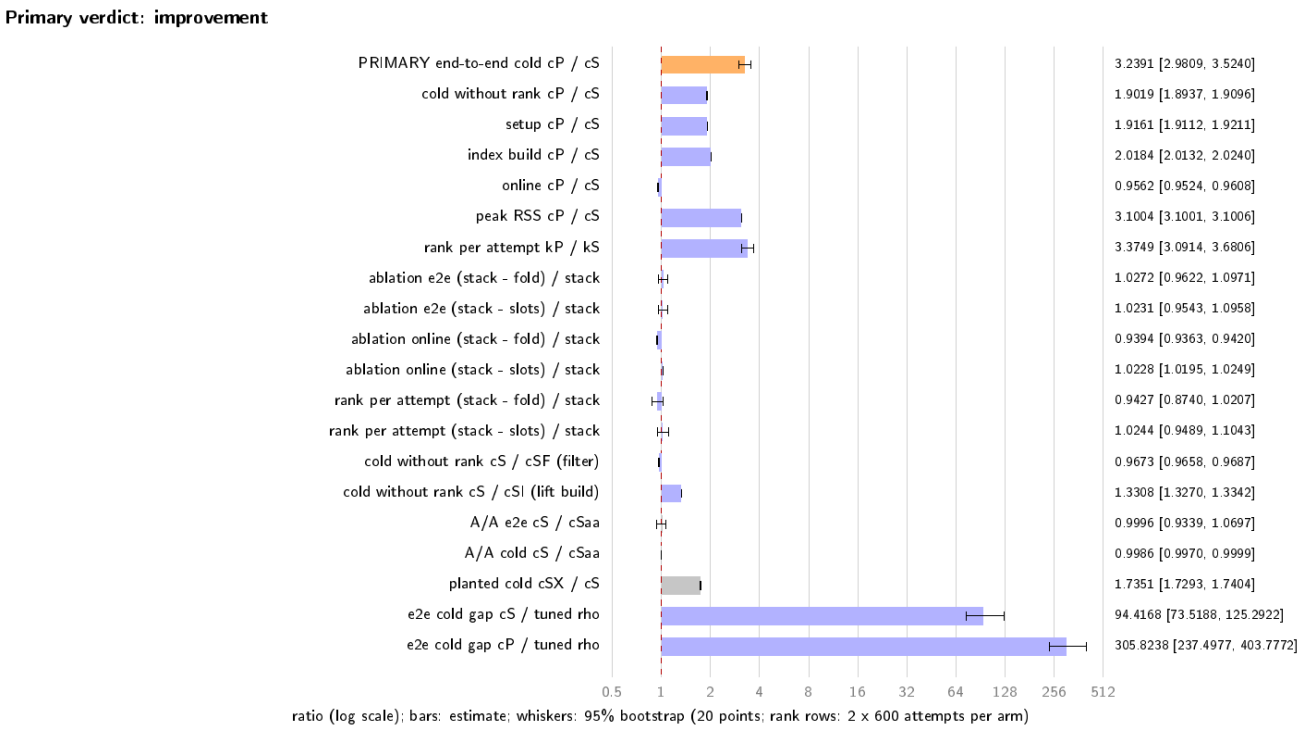


Figure 2: Frozen ratios with 95% bootstrap intervals, all from `result.json` (source, vector, rendered by `make_figures.py`).

Primary (frozen rule)

Improvement iff the 95% lower bound of R = E2E(kP, cP) / E2E(kS, cS) exceeds 1.

Statistic	Estimate [95%]	Reading
R = E2E cP / cS (20 points, 2 + 2 rank processes)	**3.2391 [2.9809, 3.5240]**	**improvement**
A/A E2E cS / cSaa	0.9996 [0.9339, 1.0697]	pass (inside [0.97, 1.03])
A/A cold cS / cSaa	0.9986 [0.9970, 0.9999]	pass (inside [0.97, 1.03])
planted cold cSX / cS	1.7351 [1.7293, 1.7404]	pass (lower bound above 1)

End-to-end cold means (rank stage mean over two processes + mean cold over 20 points): cP 869.8 + 49.4 = 919.2 s; cS 257.8 + 26.0 = 283.8 s. Single-process end-to-end cold ratio kP / kS 3.2397.

Secondaries (none changes the verdict)

Quantity	cP (baseline)	cS (stack)	Ratio cP / cS [95%]
cold without rank stage (ms)	49,372.8	25,960.4	1.9019 [1.8937, 1.9096]
setup (ms)	49,004.5	25,575.2	1.9161 [1.9112, 1.9211]
index build (ms)	46,404.8	22,991.2	2.0184 [2.0132, 2.0240]
online (ms)	368.3	385.1	0.9562 [0.9524, 0.9608] (geomeans 0.9580 [0.9533, 0.9622])
peak RSS (MiB)	5,544.6	1,788.4	3.1004 [3.1001, 3.1006]
rank stage, mean per process (s)	869.8	257.8	3.3737
rank per attempt (ms; probes)	1,449.4; 2,480,245	429.5; 1,678,903	3.3749 [3.0914, 3.6806]

- **Online is not faster; it is about 4.6% slower** in the stack (cS online 385.1 ms against 368.3 ms). The ablations below place the cost in the fold (about 6.4% online); the 16-byte slots give back about 2.3%. The gain is in the rank stage and setup.
- **Ablations (the stack minus one part; mean per-arm numbers).**

Ablation	E2E ratio [95%]	online	cold without rank	setup	rank per attempt
stack minus fold (`cSmF`, `kSmF`) / stack	1.0272 [0.9622, 1.0971]	0.9394 [0.9363, 0.9420] (geomean 0.9396)	1.8662 [1.8594, 1.8722]	1.8802 [1.8782, 1.8820]	0.9427 [0.8740, 1.0207]
stack minus slots (`cSmS`, `kSmS`) / stack	1.0231 [0.9543, 1.0958]	1.0228 [1.0195, 1.0249] (geomean 1.0217)	1.0106 [1.0090, 1.0120]	1.0104 [1.0088, 1.0118]	1.0244 [0.9489, 1.1043]

  End to end, neither ablation is separated from 1 (both intervals contain 1): the attempt-level rank-stage intervals are wide, and the rank stage dominates the end-to-end cost. The fold's effect shows in the setup (1.88x) and in a 6.4% online cost; the slots' effect shows in the 2.4% online and 1% setup gains and in peak RSS (the 32-byte table without slots peaks at 2,812 MiB against 1,788 MiB; without the fold 3,497 MiB).

- **Optional components.** The 32 MiB filter (`cSF`) is slower cold without rank (cS / cSF 0.9673 [0.9658, 0.9687]) and 7.2% faster online (online cS / cSF 1.0715 [1.0684, 1.0746]); lift_scan in the index build (`cSI`) is 1.3308 [1.3270, 1.3342] faster cold without rank, with online unchanged (cS / cSI 1.0013 [0.9980, 1.0047]). Neither is in the stack; `cSI` is a candidate next improvement to the setup.

- **S** (cost over this run's measured rT770 step, 316.66 ns, per sqrt(r)):

Arm	E2E cold	cold without rank	online
cP	31.356	1.6843	0.012562
cS	9.680	0.8856	0.013137
cSaa	9.684	0.8869	0.013115
cSmF	9.944	1.6527	0.012341
cSmS	9.904	0.8949	0.013437
cSF, cSI, cSX	not E2E	0.9156, 0.6655, 1.5366	0.012260, 0.013120, 0.013120

  Delta S end-to-end cold cP to cS = -21.675; delta S online +0.000575. The tuned rho's own `S` here: cold 0.10253, online 0.09443.

- **Gap to tuned rho rT770** (IC / rho; 20 points; distance remaining, not a gate). End-to-end cold: cP 305.8238 [237.4977, 403.7772], cS 94.4168 [73.5188, 125.2922]. Online: tuned rho / cP 7.5168 [4.4063, 13.5984], tuned rho / cS 7.1879 [4.1981, 12.9440] (IC / rho 0.1391 [0.0773, 0.2368] for cS). Mean rT770 cold 3,005.6 ms, online 2,768.1 ms.
- **Rank stage.** Per-attempt cost: kP / kS 3.3749 [3.0914, 3.6806]; kS / kSaa 0.9997 [0.9266, 1.0741]; kSmF / kS 0.9427 [0.8740, 1.0207]; kSmS / kS 1.0244 [0.9489, 1.1043]. Whole-stage ratios kP / kS 3.3737, kSmF / kS 0.9427, kSmS / kS 1.0244. Mean attempt probes: kP 2,480,245; kS, kSaa, kSmS 1,678,903; kSmF 1,679,182 (the unfolded scan's probes). After the stack the rank stage is still 91% of the end-to-end cold cost (257.8 of 283.8 s).

Limits of this result

- Tier 1, instance scope: one runner (`runnervm8k3b6`), one pinned CPU, 20 cold points, two rank processes per rank arm, one fresh process per arm per point. The primary's interval is over the points and the rank attempts of those two processes (the rank blocks were not repeated more often).
- The power analysis resolves effects of about 15% and larger end to end; the ablations (about 2-3%) are not separated end to end, and the sign of the rank-stage ablation effects is within noise. They are secondaries.
- The stack is a combination of five promoted steps, each on its own parent. The primary compares it with pP, which has B's rank stage; the 3.24x is the combined gain of the fold, the 16-byte slots and the rank-stage change (lift_scan, Feistel and the stack's probe loop on the rank stage), not of one part.
- The end-to-end cold figure is the first measured n = 83 cold `S` after B's baseline (32.93 in B's run, a different run, instance set and step time). The gap to the tuned rho is still about 94x.

Files

File	Role
`protocol.json`, `make_protocol.py`, `schedule.json`, `make_schedule.py`	frozen protocol, its builder, frozen run order
`make_online_source.py`, `make_tablefilter_source.py`, `make_probepipe_source.py`, `make_combined_source.py`, `online/`, `lift2/`	the stacked producer (verbatim parent derivations, digest-checked, plus the fold port and rank selectors); tuned rho; the two-limb canonicalizer crate
`restore.py`, `check_log_table.py`	restore and digest-check crypto 770f6a69 and every local source; re-verify B's factor-log table
`make_targets.py`, `inputs/cold_targets.json`	the new cold panel
`inputs/archived_identities.json`	archived Q0 identity (from `koblitz-n83-tablefilter-20261015`)
`make_power_prior.py`, `inputs/power_prior.json`, `power_check.py`, `analysis/power.json`	power prior, simulation, result
`arm.py`, `run_schedule.py`	one arm per pinned process; the schedule runner

File	Role
verify_receipt.py	gates, statistics, decision rule; result.json, result_table.md
figures/design.*	design diagram (TikZ source, PDF, PNG)
smoke/	local diagnostics
run_github_38862448935/	the archived artifact: allocation, builds, raw runs, panel/result.json
make_figures.py, figures/result.*	result figure (TikZ source, PDF, PNG)
README.pdf	this report: pandocREADME.md-oREADME.pdf--pdf-engine=pdflatex-Hreport_header.tex--lua-filterreport_filter.lua-Vgeometry:margin=2cm-Vfontsize=10pt-Vcolorlinks=true